Exercise Sheet 4

Centrality and Structural Roles

5942UE Network Science

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4.A Centrality Measures

Exercise 4.A.1: Centrality by Hand

Consider the graph with nodes A, B, C, D, E, F and edges A–B, A–C, B–C, B–D, D–E, D–F, E–F.

1. Compute **degree centrality** $C_d(v) = \deg(v)/(|V| - 1)$ for all nodes.
2. Identify the shortest paths between all node pairs. Which node lies on the most shortest paths between other nodes? (Compute betweenness centrality by inspection.)
3. Which node has the highest degree centrality? Which has the highest betweenness centrality? Are they the same node?
4. Interpret: what structural role does the node with the highest betweenness play in this network?

Solution:

1. Degree centrality ($|V| - 1 = 5$):

Node	Degree	C_d
A	2	2/5
B	3	3/5
C	2	2/5
D	3	3/5
E	2	2/5
F	2	2/5

2. Betweenness — by inspection:

The graph has two triangle-like clusters: A, B, C on the left and D, E, F on the right, joined by edge B–D. Every shortest path from any node in A, C to any node in D, E, F must pass through B

then D. Both B and D have very high betweenness; D slightly higher because it is the unique gateway to E, F.

3. Highest degree centrality: B and D are tied at $3/5$. **Highest betweenness:** D (and B close behind), but D edges to E and F which are a "terminal" cluster, making D the unique gateway.

4. Structural role of D: D is a **broker** or **bridge** node. It sits on the unique path connecting the left cluster A, B, C to the right cluster D, E, F. Removing D would disconnect those two groups. High betweenness with moderate degree is the signature of a structural bridge.

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Exercise 4.A.2: When Centrality Measures Disagree

Using `nx.karate_club_graph()`:

1. Compute degree, betweenness, closeness, and eigenvector centrality for all nodes.
2. Find the top-3 nodes for each measure. Do the top lists overlap?
3. Plot four subplots, each showing the network with node sizes proportional to one centrality measure.
4. Find a node that ranks **very differently** across measures. What does this reveal about its structural position?

Solution:

```python
import networkx as nx
import matplotlib.pyplot as plt

G = nx.karate_club_graph()

# Centrality dashboard
centralities = {
    "Degree":      nx.degree_centrality(G),
    "Betweenness": nx.betweenness_centrality(G),
    "Closeness":   nx.closeness_centrality(G),
    "Eigenvector": nx.eigenvector_centrality(G),
}

# Top-3 per measure
for name, c in centralities.items():
    top3 = sorted(c, key=c.get, reverse=True)[:3]
    print(f"{name}: {top3}")
```

Interpretation:

- **Leading hubs:** Nodes 0 and 33 lead in almost every measure — they are the faction leaders with high degree and influence.
- **Structural disagreement:** Node 32 often shows high betweenness relative to its degree because it acts as a bridge to node 33's faction. High eigenvector centrality in dense cores often pairs with low betweenness due to redundant paths.

4.B Structural Roles

Exercise 4.B.1: Embedded vs. Broker

Consider a graph with two triangles $1, 2, 3$ and $4, 5, 6$ connected by edge 3–4, plus node 7 connected only to nodes 3 and 4.

1. Is node 1 **embedded** or a **broker**? What evidence supports your answer?
2. Is node 7 **embedded** or a **broker**? What is its clustering coefficient C_7?
3. If edge 3–4 is removed, what happens to node 7's structural role?
4. Burt (1992) argues brokers gain "information advantage." What unique information could node 7 access that nodes in $1, 2, 3$ cannot?

Solution:

1. Node 1 — embedded: Node 1 is surrounded by the closed triangle $1, 2, 3$ where edges 1–2, 1–3, 2–3 all exist. Its neighbours are mutually connected. Clustering coefficient $C_1 = 1$. Node 1 cannot reach $4, 5, 6$ without going through 3 (and then 4), so it has no direct brokerage opportunity.

2. Node 7 — broker: Node 7 connects to 3 and 4, which are the two bridge points between the triangles. Are 3 and 4 connected? Yes — edge 3–4 exists. So $C_7 = 1$ (the one possible edge between 7's two neighbours exists). Despite $C_7 = 1$, node 7 structurally straddles two groups: it can hear information from both the $1, 2, 3$ cluster (via 3) and the $4, 5, 6$ cluster (via 4).

3. If 3–4 is removed: Node 7 would then connect to 3 (in cluster $1, 2, 3$) and 4 (in cluster $4, 5, 6$) with no edge between its neighbours → $C_7 = 0$. Now node 7 becomes the **only** bridge between the two components. Its brokerage role becomes critical.

4. Information advantage: Node 7 can learn about jobs, ideas, and events in the $4, 5, 6$ cluster that are invisible to $1, 2, 3$ (and vice versa), because information circulates within dense clusters but rarely

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crosses to the other side without a broker. This is Burt's "structural hole" argument: the gap between the two triangles is the structural hole, and 7 bridges it.

Exercise 4.B.2: Betweenness as a Proxy for Brokerage

Using the two-triangle + node 7 graph from Exercise 4.B.1:

1. Build the graph in NetworkX. Compute betweenness centrality for all nodes.
2. Verify that node 7 has high betweenness despite $C_7 = 1$.
3. Explain the apparent contradiction: high clustering coefficient yet high betweenness.
4. On the karate club graph: find the 3 nodes with the highest betweenness but **lowest** clustering coefficient. What does this combination mean structurally?

Solution:

```python
import networkx as nx

G = nx.Graph()
G.add_edges_from([(1,2),(2,3),(1,3), (4,5),(5,6),(4,6), (3,4), (7,3),(7,4)])

bc = nx.betweenness_centrality(G, normalized=True)
cc = nx.clustering(G)

# High betweenness despite high clustering coefficient
# Score = betweenness / (clustering + 1e-6) — ranks bridge-brokers
K = nx.karate_club_graph()
bc_k = nx.betweenness_centrality(K)
cc_k = nx.clustering(K)
scores = {n: bc_k[n] / (cc_k[n] + 1e-6) for n in K.nodes()}
top3 = sorted(scores, key=scores.get, reverse=True)[:3]
print("Bridge-brokers:", top3)
```

Interpretation:

- **Metric contrast:** Clustering coefficient is a local density measure (neighbours connected?), while betweenness is a global routing measure (on shortest paths?).
- **Brokerage with high C:** Node 7 has $C_7 = 1$ because its two neighbours are tied, but it still has high betweenness because it provides an alternative path between two large clusters.
- **Pure bridges:** High betweenness with low clustering (like node 32 in the karate club) indicates a pure bridge connecting otherwise separate groups.